Important SQLs for DBAs

1. Get SQL Performance Detail for a Given SQL ID

Retrieves performance metrics for a currently running or recently completed SQL (must be available in Shared Pool):

```
SELECT
    ROUND(a.elapsed_time/(1000*1000),2) elapsed_time,
    ROUND((a.elapsed_time/(1000*1000))/a.executions,2) elapsed_time_per_exec,
    a.sql_id,
    SUBSTR(a.sql_text, 1,40) sql_text,
    a.executions,a.users_executing,
    a.first_load_time,
    a.rows_processed,
    a.parsing_schema_name,
    a.plan_hash_value,
    a.module,
    a.last_load_time
FROM v$sqlarea a
WHERE a.executions > 0
  and sql_id='g306ta1yzcfuq';
```

1a. Get All SQLs Currently Running or Recently Completed

If SQL is not available in v$sqlarea, check DBA_HIST_SQLSTAT view:

```
SELECT to_char(last_active_time, 'dd-mon-yyyy hh24:mi:ss') last_active_time,
    ROUND(a.elapsed_time/(1000*1000),2) elapsed_time,
    ROUND((a.elapsed_time/(1000*1000))/a.executions,2) elapsed_time_per_exec,
    a.sql_id,
    SUBSTR(a.sql_text, 1,40) sql_text,
    a.executions,a.users_executing,
    a.first_load_time,
    a.rows_processed,
    a.parsing_schema_name,
    a.plan_hash_value,
    a.module,
    a.last_load_time
FROM v$sqlarea a
WHERE a.executions > 0
 and parsing_schema_name NOT IN ('SYS','SYSTEM','DBSNMP')
 order by last_active_time;
```

2. Get All Expensive SQLs

Identifies resource-intensive queries currently running or recently completed:

```
select
ROUND(a.elapsed_time/(1000*1000),2) elapsed_time,
ROUND((a.elapsed_time/(1000*1000))/a.executions,2) elapsed_time_per_exec,
a.sql_id,
SUBSTR(a.sql_text, 1,40) sql_text,
a.executions,a.users_executing,
a.first_load_time,
a.rows_processed,
a.parsing_schema_name,
a.plan_hash_value,
a.module,
a.last_load_time , a.*
FROM v$sqlarea a
WHERE a.executions > 100
and ROUND((a.elapsed_time/(1000*1000))/a.executions,2) > 0
order by ROUND((a.elapsed_time/(1000*1000))/a.executions,2) desc , a.executions desc;
```

3. Generate Kill Session Command for a Single User

```
set lines 300
select 'alter system kill session '||''''||s.sid||','||s.serial#||''' immediate;' kill_command,
       s.sid, s.serial#, s.status, p.spid
 from v$session s, v$process p
 where s.username = '&username'
and p.addr (+) = s.paddr;
```

4. Check Archive Destination Status

For multiple Archive destinations (Local or Standby):

```
set lines 300
col dest_name format A20
select dest_id, dest_name, error from v$archive_dest_status;
```

5. Get Database Information

```
select current_scn, name, open_mode, database_role, flashback_on from v$database;
```

6. Get Redo Generation Rate Per Hour

```
select to_char(completion_time, 'DD-MON-YYYY HH24') DAY, Thread#,
count(*) "Archive File Count", round(sum(blocks*block_size)/1024/1024/1024) GB
from v$archived_log
where completion_time between sysdate-1 and sysdate
group by to_char(completion_time, 'DD-MON-YYYY HH24'), thread# order by 1,2;
```

7. Get Redo Generation Rate Per Day

```
select to_char(completion_time, 'DD-MON-YYYY') DAY, Thread#,
 count(*) "Archive File Count", round(sum(blocks*block_size)/1024/1024/1024) GB
 from v$archived_log
 where completion_time > (sysdate - 7)
 group by to_char(completion_time, 'DD-MON-YYYY'), thread# order by 1,2;
```

8. Get List of Child Tables for a Given Table

```
SET LINES 300
col parent_owner format A15
col child_owner format A15
col parent_table format A30
col child_table format A30
col parent_cons_name format A30
col child_cons_name format A30

SELECT
      p.owner parent_owner,
      c.owner child_owner,
      p.table_name Parent_table,
      p.constraint_name parent_cons_name,
      c.table_name Child_table,
      c.constraint_name child_cons_name
  FROM
      dba_constraints p, dba_constraints c
where p.owner='&parent_Owner'
  and p.table_name='&Parent_Table_Name'
  and p.constraint_type IN ('P','U')
  and c.constraint_type='R'
  and p.owner=c.r_owner
  and p.constraint_name=c.r_constraint_name
order by 4;
```

9. Get List of Parent Table for a Given Child Table

```
SET LINES 300
col parent_owner format A15
col child_owner format A15
col parent_table format A30
col child_table format A30
col parent_cons_name format A30
col child_cons_name format A30

SELECT
      p.owner parent_owner,
      c.owner child_owner,
      p.table_name Parent_table,
      p.constraint_name parent_cons_name,
      c.table_name Child_table,
      c.constraint_name child_cons_name
  FROM
      dba_constraints p, dba_constraints c
where c.owner='&child_Owner'
  and c.table_name='&Child_Table_Name'
  and p.constraint_type IN ('P','U')
  and c.constraint_type='R'
  and p.owner=c.r_owner
  and p.constraint_name=c.r_constraint_name
order by 4;
```

10. Get Instance Detail

```
set lines 300
col host_name format A25
alter session set nls_date_format='dd-mon-yyyy hh24:mi:ss';

select instance_number, instance_name, host_name, startup_time, status from v$instance;
```

11. Get Your Session Detail

```
col Database Format A10
col Username Format A10
col current_schema format A10
SELECT
 sysdate,
 sys_context('USERENV','SERVER_HOST') server_host,
 sys_context('USERENV','SERVICE_NAME') service_name,
 sys_context('USERENV','INSTANCE_NAME') instance_name,
 sys_context('USERENV','DB_UNIQUE_NAME') DB_UNIQUE_NAME,
 sys_context('USERENV','DB_NAME') DB_NAME,
 sys_context('USERENV','DB_DOMAIN') DB_DOMAIN,
 sys_context('USERENV','SESSION_USER') Username,
 sys_context('USERENV','CURRENT_SCHEMA') current_schema
 FROM Dual;
```

12. Find Plan Changes for a SQL ID

Changes in plan_hash_value column indicate a change in execution plan:

```
select
  sn.begin_interval_time,
  sn.end_interval_time,
  sq.sql_id,
  sq.plan_hash_value,
  sq.executions_total,
  sq.executions_delta,
  round(sq.elapsed_time_total/(1000*1000),3) elapsed_time_total_in_sec,
  round(sq.elapsed_time_delta/(1000*1000),3) elapsed_time_delta_in_sec,
  sq.module,
  sq.action,
  round(sq.cpu_time_total/(1000*1000),3) cpu_time_total_in_sec,
  round(sq.cpu_time_delta/(1000*1000),3) cpu_time_delta_in_sec
from dba_hist_sqlstat sq, dba_hist_snapshot sn
where sq.sql_id='gsd2mfhfcynwg'
and sq.snap_id = sn.snap_id
order by sn.end_interval_time;
```

13. Get Bind Variable Values for a SQL ID

```
select * from v$sql_bind_capture where sql_id='gsd2mfhfcynwg';
```

14. Get Plan History Detail for a SQL ID

```
select * from dba_hist_sql_plan sq where sq.sql_id='gsd2mfhfcynwg';
```

15. Get List of SQL Baselines

```
select * from dba_sql_plan_baselines;
```

16. Find SQL Executed by OS Process ID

```
prompt "Please Enter The UNIX Process ID"
set pagesize 50000
set linesize 30000
set long 500000
set head off
select
s.username su,
substr(sa.sql_text,1,540) txt
from v$process p,
v$session s,
v$sqlarea sa
where p.addr=s.paddr
and s.username is not null
and s.sql_address=sa.address(+)
and s.sql_hash_value=sa.hash_value(+)
and spid=&SPID;
```

17. List Sessions Based on CPU Usage

```
set pages 1000
set lines 1000
col OSPID for a06
col SID for 99999
col SERIAL# for 999999
col SQL_ID for a14
col USERNAME for a15
col PROGRAM for a23
col MODULE for a18
col OSUSER for a10
col MACHINE for a25
select * from (
select p.spid "ospid",
se.SID,ss.serial#,ss.SQL_ID,ss.username,substr(ss.program,1,22) "program",
ss.module,ss.osuser,ss.MACHINE,
se.VALUE/100 cpu_usage_sec
from
v$session ss,
v$sesstat se,
v$statname sn,
v$process p
where
se.STATISTIC# = sn.STATISTIC#
and
NAME like '%CPU used by this session%'
and
se.SID = ss.SID
and ss.status='ACTIVE'
and ss.username is not null
and ss.paddr=p.addr
order by se.VALUE desc);
```

18. Session Detail

```
select
    a.event,
    a.sid,
    a.username,
    a.blocking_session,
    a.final_blocking_session,
    a.sql_id,
    a.status,
    a.osuser,
    a.machine,
    a.program,
    a.last_call_et,
    a.blocking_session_status,
    a.final_blocking_session_status,
    a.wait_class,
    a.seconds_in_wait,
    a.state
FROM
  v$session a where username is not null
and event not in ('SQL*Net message from client','SQL*Net message to client');
```

19. SQL Detail from V$SQL

```
select
 sql_id,
 plan_hash_value,
 first_load_time,
 executions,
 round(elapsed_time/(1000*1000),3) elapsed_time_total_in_sec
 from V$sql
where sql_id IN ('fsy3ubymdz500','g306ta1yzcfuq');
```
